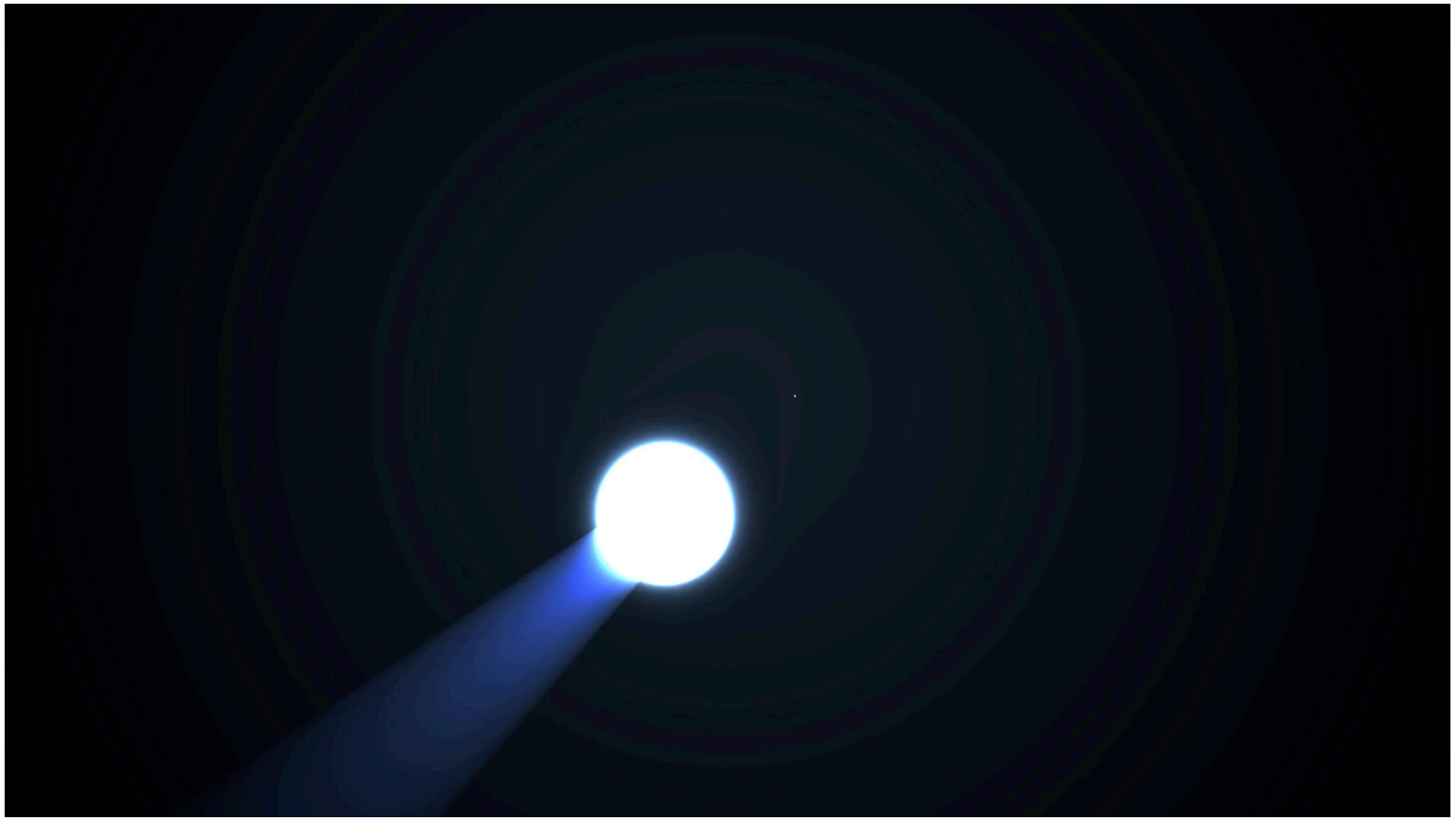
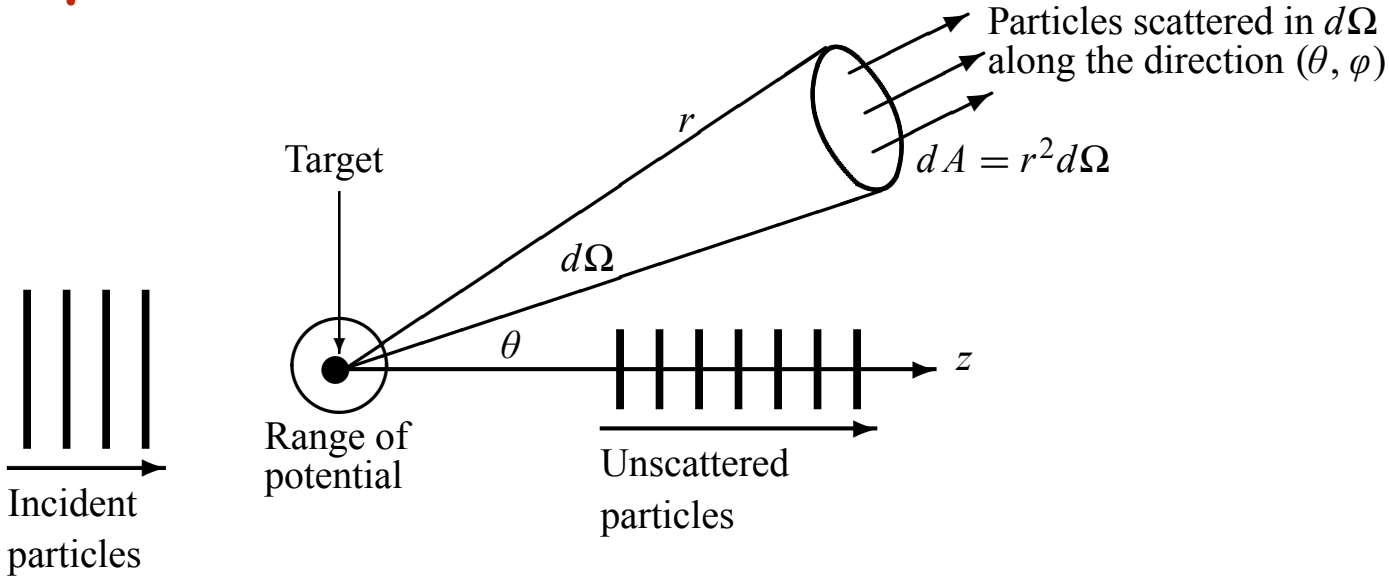
Quantum mechanics

Chapter VIII Scattering theory



In a scattering experiment, one observes the collisions between a beam of incident particles and a target material. The total number of collisions over the duration of the experiment is proportional to the total number of incident particles and to the number of target particles per unit area in the path of the beam.



The number of particles scattered into an element of solid angle

$$d\Omega = \sin\theta \, d\theta \, d\varphi$$

is proportional to a quantity that plays a central role in the physics of scattering: the *differential cross section*. The differential cross section, which is is defined as the number of particles scattered into an element of solid angle $d\Omega$ in the direction (θ,φ) per unit time and incident flux:

$$\frac{d\sigma(\theta,\varphi)}{d\Omega} = \frac{1}{J_{inc}} \frac{dN(\theta,\varphi)}{d\Omega},$$

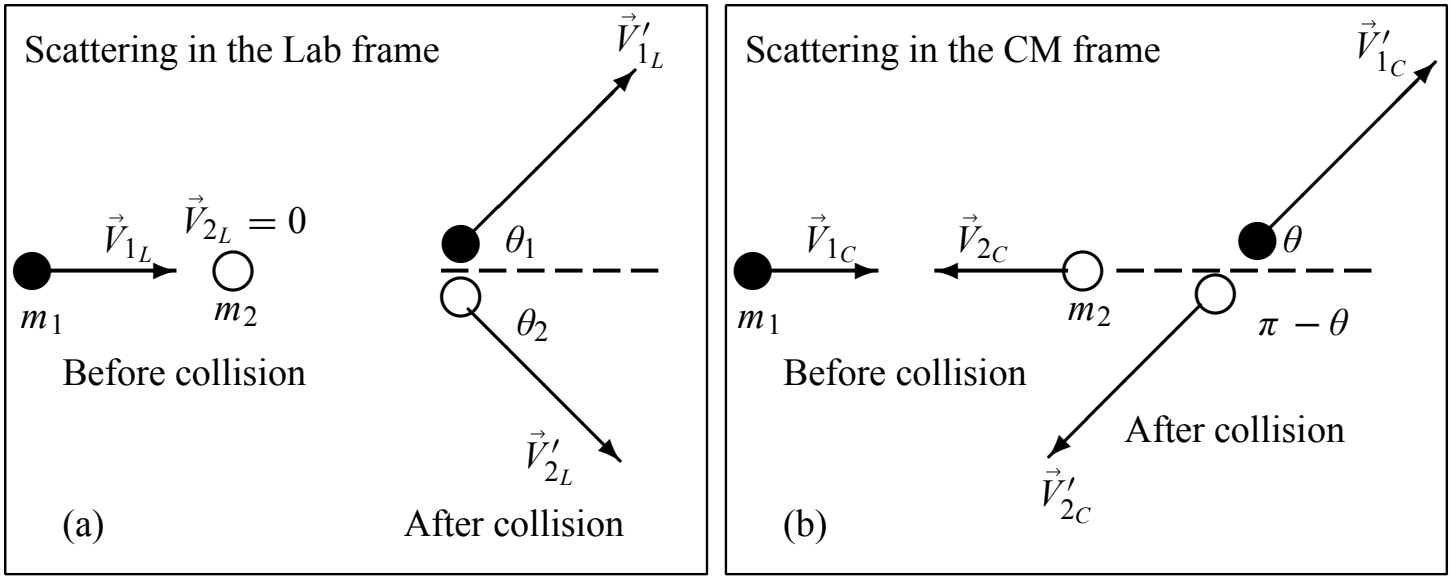
where J_{inc} is the incident flux (or incident current density); it is equal to the number of incident particles per area per unit time.

The relationship between $d\sigma/d\Omega$ and the total cross section σ is obvious:

$$\sigma = \int \frac{d\sigma}{d\Omega} d\Omega = \int_0^{\pi} \sin\theta \, d\theta \int_0^{2\pi} \frac{d\sigma(\theta, \varphi)}{d\Omega} d\varphi .$$

Most scattering experiments are carried out in the laboratory (Lab) frame in which the target is initially at rest while the projectiles are moving. Calculations of the cross sections are generally easier to perform within the center of mass (CM) frame in which the center of mass of the projectiles-target system is at rest (before and after collision).

To find the connection between the Lab and CM cross sections, we need first to find how the scattering angles in one frame are related to their counterparts in the other. Let us consider the scattering of two particles of masses m_1 and m_2; m_2 represents the target, which is initially at rest, and m_1 the projectile.



A scattering happens in the Lab and CM frames, where θ_1 and θ are the scattering angles of m_1 in the Lab and CM frames, respectively; we are interested in detecting m_1. There is

$$\vec{V}_{1_L} = \vec{V}_{1_C} + \vec{V}_{CM},$$

where V_{1_L} and V_{1_C} are the velocities of m_1 in the Lab and CM frames *before* collision and V_{CM} is the velocity of the CM with respect to the Lab frame. Similarly, the velocity of m_1 *after* collision is

$$\vec{V}'_{1_L} = \vec{V}'_{1_C} + \vec{V}_{CM}.$$

We can infer the x and y components in above equation

$$V'_{1_L}\cos\theta_1 = V'_{1_C}\cos\theta + V_{CM},$$
$$V'_{1_L}\sin\theta_1 = V'_{1_C}\sin\theta.$$

Therefore,

$$\tan\theta_1 = \frac{\sin\theta}{\cos\theta + V_{CM}/V'_{1_C}},$$

To evaluate V_{cm}/V'_{1c}, we have

$$\vec{V}_{CM} = \frac{m_1\vec{V}_{1_L} + m_2\vec{V}_{2_L}}{m_1+m_2} = \frac{m_1}{m_1+m_2}\vec{V}_{1_L},$$

therefore,

$$\vec{V}_{1_C} = \left(1 - \frac{m_1}{m_1+m_2}\right)\vec{V}_{1_L} = \frac{m_2}{m_1+m_2}\vec{V}_{1_L}.$$

On the other hand, since the center of mass is at rest in the CM frame, the total momenta before and after collisions are separately zero:

$$p_C = m_1 V_{1C} - m_2 V_{2C} = 0 \quad \Longrightarrow \quad V_{2C} = \frac{m_1}{m_2} V_{1C},$$

$$p'_{C_x} = m_1 V'_{1C} \cos\theta - m_2 V'_{2C} \cos\theta = 0 \quad \Longrightarrow \quad V'_{2C} = \frac{m_1}{m_2} V'_{1C}.$$

In the case of *elastic* collision, the speeds of the particles in the CM frame are the same before and after collision since the kinetic energy is conserved,

$$\tfrac{1}{2} m_1 V_{1C}^2 + \tfrac{1}{2} m_2 V_{2C}^2 = \tfrac{1}{2} m_1 {V'_{1C}}^2 + \tfrac{1}{2} m_2 {V'_{2C}}^2$$

Thus, we can rewrite

$$\vec{V}'_{1C} = \vec{V}_{1C} = \frac{m_2}{m_1 + m_2}\vec{V}_{1L}.$$

Finally we have

$$\frac{V_{CM}}{V'_{1C}} = \frac{m_1}{m_2}.$$

Hence

$$\tan\theta_1 = \frac{\sin\theta}{\cos\theta + V_{2C}/V_{1C}} = \frac{\sin\theta}{\cos\theta + m_1/m_2},$$

And

$$\cos\theta_1 = \frac{\cos\theta + \frac{m_1}{m_2}}{\sqrt{1 + \frac{m_1^2}{m_2^2} + 2\frac{m_1}{m_2}\cos\theta}}.$$

Similarly

$$\tan\theta_2 = \frac{\sin\theta}{-\cos\theta + V_{CM}/V'_{2C}} = \frac{\sin\theta}{1 - \cos\theta} = \cot\left(\frac{\theta}{2}\right) \Longrightarrow \theta_2 = \frac{\pi - \theta}{2}.$$

The connection between the differential cross sections in the Lab and CM frames can be obtained from the fact that the number of scattered particles passing through an infinitesimal cross section $d\sigma$ is the same in both frames: $d(\theta_1,\varphi_1)=d(\theta,\varphi)$. What differs is the solid angle $d\Omega$, since it is given in the Lab frame and in the CM frame by

$$d\Omega_1 = \sin\theta_1 d\theta_1 d\varphi_1 \qquad d\Omega = \sin\theta\, d\theta d\varphi$$

Thus, we have

$$\left(\frac{d\sigma}{d\Omega_1}\right)_{Lab} d\Omega_1 = \left(\frac{d\sigma}{d\Omega}\right)_{CM} d\Omega \Longrightarrow \left(\frac{d\sigma}{d\Omega_1}\right)_{Lab} = \left(\frac{d\sigma}{d\Omega}\right)_{CM} \frac{\sin\theta}{\sin\theta_1}\frac{d\theta}{d\theta_1}\frac{d\varphi}{d\varphi_1},$$

Since there is cylindrical symmetry around the direction of the incident beam, we have

$$\varphi = \varphi_1$$

Hence, we have

$$\left(\frac{d\sigma}{d\Omega_1}\right)_{Lab} = \left(\frac{d\sigma}{d\Omega}\right)_{CM} \frac{d(\cos\theta)}{d(\cos\theta_1)}.$$

and

$$\left(\frac{d\sigma}{d\Omega_1}\right)_{Lab} = \frac{(1+\frac{m_1^2}{m_2^2}+2\frac{m_1}{m_2}\cos\theta)^{3/2}}{1+\frac{m_1}{m_2}\cos\theta}\left(\frac{d\sigma}{d\Omega}\right)_{CM}.$$

Similarly

$$\left(\frac{d\sigma}{d\Omega_2}\right)_{Lab} = 4\cos\theta_2\left(\frac{d\sigma}{d\Omega_2}\right)_{CM} = 4\sin\left(\frac{\theta}{2}\right)\left(\frac{d\sigma}{d\Omega_2}\right)_{CM}$$

For simplicity, we consider the case of elastic scattering between two spinless, nonrelativistic particles of masses m_1 and m_2. If the interaction is *time independent*, we can describe the two-particle system with stationary states

$$\Psi(\vec{r}_1, \vec{r}_2, t) = \psi(\vec{r}_1, \vec{r}_2) e^{-iE_T t/\hbar},$$

where is a solution of the time-independent Schrödinger equation:

$$\left[-\frac{\hbar^2}{2m_1}\vec{\nabla}_1^2 - \frac{\hbar^2}{2m_2}\vec{\nabla}_2^2 + \hat{V}(\vec{r}_1, \vec{r}_2) \right] \psi(\vec{r}_1, \vec{r}_2) = E_T \psi(\vec{r}_1, \vec{r}_2);$$

$V(r_1 r_2)$ is the potential representing the interaction between the two particles.

In the case where the interaction between m_1 and m_2 depends only on their relative distance

$$r = |\vec{r}_1 - \vec{r}_2|$$

The Schodinger equation can be transformed to CM frame

$$-\frac{\hbar^2}{2\mu}\vec{\nabla}^2\psi(\vec{r}) + \hat{V}(r)\psi(\vec{r}) = E\psi(\vec{r})$$

where a reduced mass is defined as

$$\mu = m_1 m_2/(m_1 + m_2)$$

In quantum mechanics the incident particle is described by means of a wave packet that interacts with the target. After scattering, the wave function consists of an unscattered part propagating in the forward direction and a scattered part that propagates along some direction (θ,φ)

We assume that $V(r)$ has a *finite* range a. Thus the interaction between the particle and the potential occurs only in a limited region of space $r<a$, which is called the *range* of $V(r)$, or the scattering region. Outside the range, $r>a$, the potential vanishes, $V(r)=0$. The incident particle can be described as

$$\left(\nabla^2 + k_0^2\right)\phi_{inc}(\vec{r}) = 0,$$

where

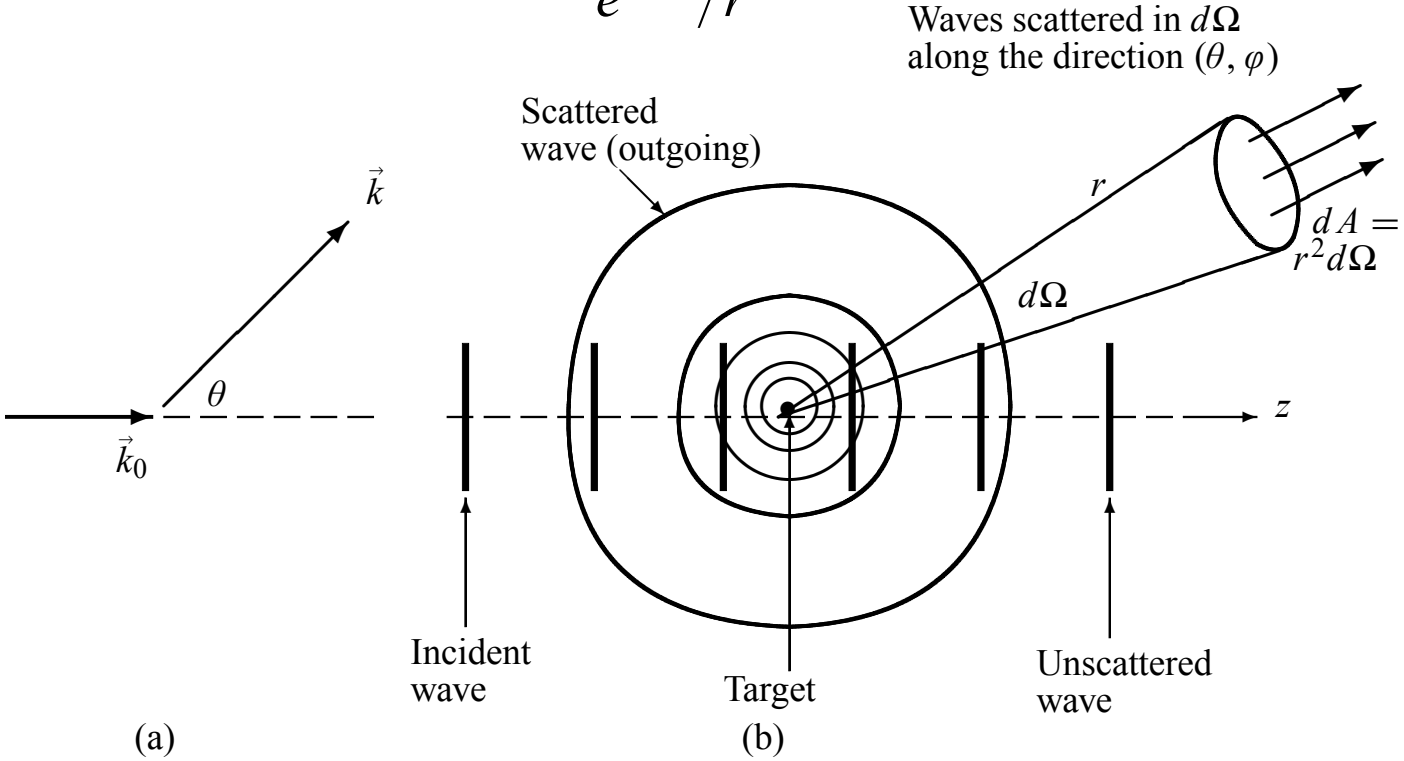
$$k_0^2 = 2\mu E/\hbar^2.$$

So, a *free* particle before collision hence can be described by a *plane* wave

$$\phi_{inc}(\vec{r}) = Ae^{i\vec{k}_0\cdot\vec{r}},$$

When the incident wave collides or interacts with the target, an outgoing wave $\psi_{sc}(r)$ is scattered out. In the case of an *isotropic* scattering, the scattered wave is *spherically*, having the form

$$e^{i\vec{k}\cdot\vec{r}}/r$$



In general, however, the scattered wave is not spherically symmetric; its amplitude depends on the direction (θ,φ) along which is detected and hence

$$\phi_{sc}(\vec{r}) = Af(\theta,\varphi)\frac{e^{i\vec{k}\cdot\vec{r}}}{r},$$

where f(θ,φ) is called the scattering amplitude. The total wave consists of a superposition of the incident plane wave and the scattered wave

$$\psi(\vec{r}) = \phi_{inc}(\vec{r}) + \phi_{sc}(\vec{r}) \simeq A\left[e^{i\vec{k}_0\cdot\vec{r}} + f(\theta,\varphi)\frac{e^{i\vec{k}\cdot\vec{r}}}{r}\right]$$

where A is a normalization factor.

Differential Cross Section

To find the relation between the scattering amplitude and differential cross section, let us first introduce the incident and scattered flux densities:

$$\vec{J}_{inc} = \frac{i\hbar}{2\mu}(\phi_{inc}\vec{\nabla}\phi^*_{inc} - \phi^*_{inc}\vec{\nabla}\phi_{inc}),$$

$$\vec{J}_{sc} = \frac{i\hbar}{2\mu}(\phi_{sc}\vec{\nabla}\phi^*_{sc} - \phi^*_{sc}\vec{\nabla}\phi_{sc}).$$

Taking the expressions of the wave functions, we end up with

$$J_{inc} = |A|^2\frac{\hbar k_0}{\mu}, \qquad J_{sc} = |A|^2\frac{\hbar k}{\mu r^2}\left|f(\theta,\varphi)\right|^2.$$

Now, we may recall that the number $dN(\theta,\varphi)$ of particles scattered into an element of solid angle

$$dN(\theta,\varphi) = J_{sc}r^2 d\Omega.$$

It yields

$$\frac{dN}{d\Omega} = J_{sc} r^2 = |A|^2 \frac{\hbar k}{\mu} \left| f(\theta, \varphi) \right|^2 .$$

Therefore, the differential cross section is

$$\frac{d\sigma}{d\Omega} = \frac{1}{J_{inc}} \frac{dN}{d\Omega} = \frac{k}{k_0} \left| f(\theta, \varphi) \right|^2 .$$

Since the normalization factor A does not contribute to the differential cross section, we will be taking it equal to one.

For elastic scattering k_0 is equal to k

$$\frac{d\sigma}{d\Omega} = \left| f(\theta, \varphi) \right|^2 .$$

Scattering Amplitude

The scattering amplitude can be obtained by solving the following Schoedinger equation

$$(\nabla^2 + k^2)\psi(\vec{r}) = \frac{2\mu}{\hbar^2} V(\vec{r})\psi(\vec{r}).$$

The general solution to this equation consists of a sum of two components: a general solution to the homogeneous equation

$$(\nabla^2 + k_0^2)\psi_{homo}(\vec{r}) = 0,$$

and a particular solution, which can express it in terms of *Green's function*. Therefore

$$\psi(\vec{r}) = \phi_{inc}(\vec{r}) + \frac{2\mu}{\hbar^2}\int G(\vec{r} - \vec{r}\,')V(\vec{r}\,')\psi(\vec{r}\,')\,d^3r',$$

where

$$\phi_{inc}(\vec{r}) = e^{i\vec{k}_0\cdot\vec{r}}$$

and the function $G(r\,r')$ is obtained by solving the point source equation

$$(\nabla^2 + k^2)G(\vec{r}-\vec{r}\,') = \delta(\vec{r}-\vec{r}\,'),$$

They are given by Fourier transformation as follows

$$G(\vec{r}-\vec{r}\,') = \frac{1}{(2\pi)^3}\int e^{i\vec{q}\cdot(\vec{r}-\vec{r}\,')}\tilde{G}(\vec{q})\,d^3q,$$

$$\delta(\vec{r}-\vec{r}\,') = \frac{1}{(2\pi)^3}\int e^{i\vec{q}\cdot(\vec{r}-\vec{r}\,')}\,d^3q.$$

Herefore

$$\left(-\vec{q}^{\,2}+\vec{k}^{\,2}\right)\tilde{G}(\vec{q}) = 1 \qquad \Longrightarrow \qquad \tilde{G}(\vec{q}) = \frac{1}{\vec{k}^{\,2}-\vec{q}^{\,2}}.$$

and

$$G(\vec{r}-\vec{r}\,') = \frac{1}{(2\pi)^3}\int \frac{e^{i\vec{q}\cdot(\vec{r}-\vec{r}\,')}}{k^2-q^2}d^3q.$$

To integrate over the angles in

$$G(\vec{r}-\vec{r}\,') = \frac{1}{(2\pi)^3}\int_0^\infty \frac{q^2dq}{k^2-q^2}\int_0^\pi e^{iq|\vec{r}-\vec{r}\,'|\cos\theta}\sin\theta\, d\theta\int_0^{2\pi} d\varphi,$$

We need simply to make the variables change $x=\cos\theta$

$$\int_0^\pi e^{iq|\vec{r}-\vec{r}\,'|\cos\theta}\sin\theta\, d\theta = \int_{-1}^1 e^{iq|\vec{r}-\vec{r}\,'|x}dx = \frac{1}{iq|\vec{r}-\vec{r}\,'|}\left(e^{iq|\vec{r}-\vec{r}\,'|}-e^{-iq|\vec{r}-\vec{r}\,'|}\right).$$

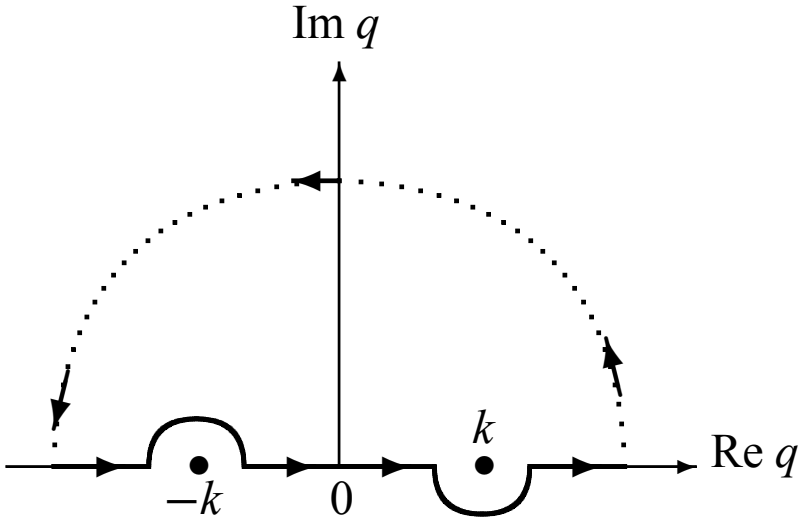
Hence

$$G(\vec{r}-\vec{r}\,') = \frac{1}{4\pi^2 i|\vec{r}-\vec{r}\,'|}\int_0^\infty \frac{q}{k^2-q^2}\left(e^{iq|\vec{r}-\vec{r}\,'|}-e^{-iq|\vec{r}-\vec{r}\,'|}\right)dq,$$

Or

$$G(\vec{r}-\vec{r}\,') = -\frac{1}{4\pi^2 i|\vec{r}-\vec{r}\,'|}\int_{-\infty}^{+\infty} \frac{qe^{iq|\vec{r}-\vec{r}\,'|}}{q^2-k^2}dq.$$

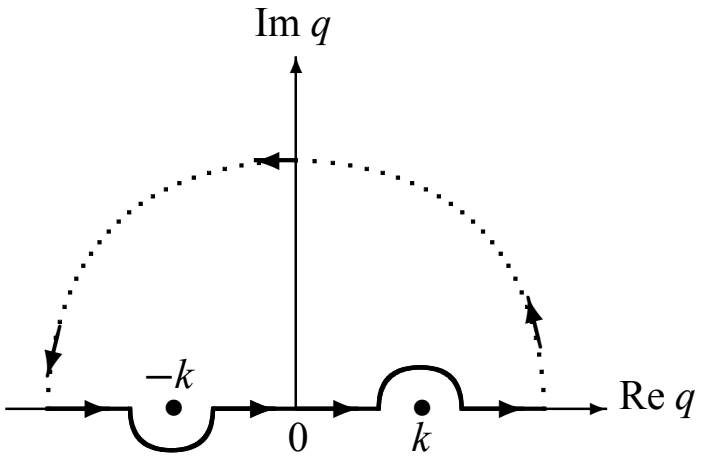
We may evaluate this integral by the method of residues by closing the contour in the upper half of the q-plane: it is equal to $2i$ times the residue of the integrand at the poles. Since there are two poles, $q=\pm k$, the integral has two possible values. The value corresponding to the pole at $q=k$, which lies inside the contour of integration in



(a) Contour for outgoing waves

$$G_+(\vec{r}-\vec{r}\,') = -\frac{1}{4\pi}\frac{e^{ik|\vec{r}-\vec{r}\,'|}}{|\vec{r}-\vec{r}\,'|}$$

and the value for the pole at $q=-k$ is



(b) Contour for incoming waves

$$G_-(\vec{r}-\vec{r}\,') = -\frac{1}{4\pi}\frac{e^{-ik|\vec{r}-\vec{r}'|}}{|\vec{r}-\vec{r}\,'|}.$$

Green's function $G_+(r,r')$ represents an outgoing spherical wave emitted from r' and the $G_-(r,r')$ corresponds to an *incoming wave* that converges onto r'. Since the scattered waves are outgoing waves, only $G_+(r,r')$ is of interest to us.

We obtain the total scattered wave function:

$$\psi(\vec{r}) = \phi_{inc}(\vec{r}) - \frac{\mu}{2\pi\hbar^2}\int \frac{e^{ik|\vec{r}-\vec{r}\,'|}}{|\vec{r}-\vec{r}\,'|} V(\vec{r}\,')\psi(\vec{r}\,')\, d^3r'.$$

This is an *integral equation*; it does not yet give the unknown solution but only contains it in the integrand. This equation can be solved approximately by means of a series of successive or iterative approximations, known as the *Born series*. The zero-order solution is given by

$$\psi_0(\vec{r}) = \phi_{inc}(\vec{r}).$$

The first-order solution is obtained

$$\begin{aligned}\psi_1(\vec{r}) &= \phi_{inc}(\vec{r}) - \frac{\mu}{2\pi\hbar^2}\int \frac{e^{ik|\vec{r}-\vec{r}_1|}}{|\vec{r}-\vec{r}_1|} V(\vec{r}_1)\psi_0(\vec{r}_1)\, d^3r_1 \\ &= \phi_{inc}(\vec{r}) - \frac{\mu}{2\pi\hbar^2}\int \frac{e^{ik|\vec{r}-\vec{r}_1|}}{|\vec{r}-\vec{r}_1|} V(\vec{r}_1)\phi_{inc}(\vec{r}_1)\, d^3r_1.\end{aligned}$$

The second order is obtained

$$\psi_2(\vec{r}) = \phi_{inc}(\vec{r}) - \frac{\mu}{2\pi\hbar^2}\int \frac{e^{ik|\vec{r}-\vec{r}_2|}}{|\vec{r}-\vec{r}_2|} V(\vec{r}_2)\psi_1(\vec{r}_2)\, d^3r_2$$

$$= \phi_{inc}(\vec{r}) - \frac{\mu}{2\pi\hbar^2}\int \frac{e^{ik|\vec{r}-\vec{r}_2|}}{|\vec{r}-\vec{r}_2|} V(\vec{r}_2)\phi_{inc}(\vec{r}_2)\, d^3r_2$$

$$+ \left(\frac{\mu}{2\pi\hbar^2}\right)^2 \int \frac{e^{ik|\vec{r}-\vec{r}_2|}}{|\vec{r}-\vec{r}_2|} V(\vec{r}_2)\, d^3r_2 \int \frac{e^{ik|\vec{r}_2-\vec{r}_1|}}{|\vec{r}_2-\vec{r}_1|} V(\vec{r}_1)\phi_{inc}(\vec{r}_1)\, d^3r_1.$$

Continuing in this way, we can obtain the wave function to the desired order.

For large values of r

$$k|\vec{r}-\vec{r}\,'| = k\sqrt{\vec{r}^2 - 2\vec{r}\cdot\vec{r}\,' + \vec{r}\,'^2} \simeq kr - k\frac{\vec{r}}{r}\cdot\vec{r}\,' = kr - \vec{k}\cdot\vec{r}\,',$$

$$\frac{1}{|\vec{r}-\vec{r}\,'|} = \frac{1}{r}\frac{1}{\left|1-\vec{r}\cdot\vec{r}\,'/\vec{r}^{\,2}\right|} \simeq \frac{1}{r}\left(1+\frac{\vec{r}\cdot\vec{r}\,'}{r^2}\right) \simeq \frac{1}{r},$$

Scattering Amplitude

The asymptotic form of total scattered wave function is

$$\psi(\vec{r}) \longrightarrow e^{i\vec{k}_0\cdot\vec{r}} + \frac{e^{ikr}}{r} f(\theta,\varphi) \qquad (r \to \infty),$$

where

$$f(\theta,\varphi) = -\frac{\mu}{2\pi\hbar^2}\int e^{-i\vec{k}.\vec{r}\,'} V(\vec{r}\,')\psi(\vec{r}\,')\, d^3r' = -\frac{\mu}{2\pi\hbar^2}\langle\phi \mid \hat{V} \mid \psi\rangle,$$

where $\phi(r)$ is a plane wave and k is the wave vector of the scattered wave. The differential cross section is then given by

$$\frac{d\sigma}{d\Omega} = |f(\theta,\varphi)|^2 = \frac{\mu^2}{4\pi^2\hbar^4}\left|\int e^{-i\vec{k}\cdot\vec{r}\,'}\hat{V}(\vec{r}\,')\psi(\vec{r}\,')\, d^3r'\right|^2 = \frac{\mu^2}{4\pi^2\hbar^4}\left|\langle\phi \mid \hat{V} \mid \psi\rangle\right|^2$$

The Born Approximation

If the potential $V(r)$ is weak enough, it will distort only slightly the incident plane wave. The *first Born approximation* consists then of approximating the scattered wave function by a *plane wave*.

$$\psi(\vec{r}) \simeq \phi_{inc}(\vec{r}) - \frac{\mu}{2\pi\hbar^2}\int \frac{e^{ik|\vec{r}-\vec{r}\,'|}}{|\vec{r}-\vec{r}\,'|} V(\vec{r}\,')\phi_{inc}(\vec{r}\,')\, d^3r'.$$

Thus, we can write the scattering amplitude and the differential cross section in the first Bonn approximation as

$$f(\theta,\varphi) = -\frac{\mu}{2\pi\hbar^2}\int e^{-i\vec{k}\cdot\vec{r}'} V(\vec{r}\,')\phi_{inc}(\vec{r}\,')\, d^3r' = -\frac{\mu}{2\pi\hbar^2}\int e^{i\vec{q}\cdot\vec{r}\,'} V(\vec{r}\,')\, d^3r',$$

$$\frac{d\sigma}{d\Omega} = \left| f(\theta,\varphi)\right|^2 = \frac{\mu^2}{4\pi^2\hbar^4}\left|\int e^{i\vec{q}\cdot\vec{r}\,'} V(\vec{r}\,')\, d^3r'\right|^2.$$

where

$$\vec{q} = \vec{k_0} - \vec{k}$$

is the momentum transfer; k_0 and k are the linear momenta of the incident and scattered particles, respectively. In *elastic scattering*, the magnitudes of k_0 and k are equal hence

$$q = \left|\vec{k_0} - \vec{k}\right| = \sqrt{k_0^2 + k^2 - 2kk_0\cos\theta} = k\sqrt{2(1-\cos^2\theta)} = 2k\sin\left(\frac{\theta}{2}\right).$$

If the potential $V(r)$ is spherically symmetric, choosing the z-axis along q

$$\begin{aligned}\int e^{i\vec{q}\cdot\vec{r}\,'} V(\vec{r}\,')\, d^3r' &= \int_0^\infty r'^2 V(r')\, dr' \int_0^\pi e^{iqr'\cos\theta'} \sin\theta' d\theta' \int_0^{2\pi} d\varphi' \\ &= 2\pi \int_0^\infty r'^2 V(\vec{r}\,')\, dr' \int_{-1}^1 e^{iqr'x} dx = \frac{4\pi}{q}\int_0^\infty r' V(r') \sin(qr')\, dr'.\end{aligned}$$

Therefore

$$f(\theta) = -\frac{2\mu}{\hbar^2 q}\int_0^\infty r' V(r')\sin(qr')\,dr',$$

and

$$\frac{d\sigma}{d\Omega} = \left|f(\theta)\right|^2 = \frac{4\mu^2}{\hbar^4 q^2}\left|\int_0^\infty r' V(r')\sin(qr')\,dr'\right|^2.$$

The first Born approximation is valid whenever the wave function is only slightly different from the incident plane wave; that is

$$\left|\frac{\mu}{2\pi\hbar^2}\int \frac{e^{ik|\vec{r}-\vec{r}\,'|}}{|\vec{r}-\vec{r}\,'|}V(r')e^{i\vec{k}_0\cdot\vec{r}\,'}\,d^3r'\right| \ll |\phi_{inc}(\vec{r})|^2. \qquad \left|\frac{\mu}{2\pi\hbar^2}\int \frac{e^{ik|\vec{r}-\vec{r}\,'|}}{|\vec{r}-\vec{r}\,'|}V(r')e^{i\vec{k}_0\cdot\vec{r}\,'}\,d^3r'\right| \ll 1.$$

In the case of a Coulomb potential,

$$V(r) = Z_1 Z_2 e^2 / r,$$

The differential cross section with Born approximation is

$$\frac{d\sigma}{d\Omega} = \frac{4 Z_1^2 Z_2^2 e^4 \mu^2}{\hbar^4 q^2} \left| \int_0^\infty \sin(qr)\, dr \right|^2,$$

where

$$\begin{aligned} \int_0^\infty \sin(qr)\, dr &= \lim_{\lambda \to 0} \int_0^\infty e^{-\lambda r} \sin(qr)\, dr = \frac{1}{2i} \lim_{\lambda \to 0} \left[\int_0^\infty e^{-(\lambda - iq)r}\, dr - \int_0^\infty e^{-(\lambda + iq)r}\, dr \right] \\ &= \frac{1}{2i} \lim_{\lambda \to 0} \left[\frac{1}{\lambda - iq} - \frac{1}{\lambda + iq} \right] = \frac{1}{q}. \end{aligned}$$

Since

$$q = 2k \sin(\theta/2)$$

Finally, the *Rutherford formula* is obtained

$$\frac{d\sigma}{d\Omega} = \left(\frac{2 Z_1 \mu Z_2 e^2}{\hbar^2 q^2} \right)^2 = \left(\frac{Z_1 Z_2 \mu e^2}{2 \hbar^2 k^2} \right)^2 \sin^{-4}\left(\frac{\theta}{2} \right) = \frac{Z_1^2 Z_2^2 e^4}{16 E^2} \sin^{-4}\left(\frac{\theta}{2} \right),$$

We can then examine how each of the partial wave is distorted by $V(r)$ after the particle scatters from the potential. The most general solution of the Schroedinger equation with a central potential is

$$\psi(\vec{r}) = \sum_{lm} C_{lm} R_{kl}(r) Y_{lm}(\theta, \varphi).$$

Since $V(r)$ is central, the system is symmetrical (rotationally invariant) about the z-axis. The scattered wave function must not then depend on the azimuthal angle, hence m=0 and

$$\psi(r, \theta) = \sum_{l=0}^{\infty} a_l R_{kl}(r) P_l(\cos\theta),$$

Each term in this equation, which is known as a *partial wave*, is a joint eigenfunction of L and L_z

The radial wave function $R_{kl}(r)$ obeys the following radial equation

$$\left[\frac{d^2}{dr^2}+k^2-\frac{l(l+1)}{r^2}\right](rR_{kl}(r))=\frac{2m}{\hbar^2}V(r)\,(rR_{kl}(r))\,.$$

On the other hand, the scattering wave function can between as

$$\psi(r,\theta)\simeq\sum_{l=0}^{\infty}i^l(2l+1)j_l(kr)P_l(\cos\theta)+f(\theta)\frac{e^{ikr}}{r}.$$

By establishing a connection between the asymptotic forms of the function and the form solution, we can determine the scattering amplitude and hence the differential cross section.

First, since the limit of the Bessel function $j_l(kr)$ for large values of r is given by

$$j_l(kr) \longrightarrow \frac{\sin(kr - l\pi/2)}{kr} \qquad (r \longrightarrow \infty),$$

The asymptotic form of wave function becomes

$$\psi(r,\theta) \longrightarrow \sum_{l=0}^{\infty} i^l (2l+1) P_l(\cos\theta) \frac{\sin(kr - l\pi/2)}{kr} + f(\theta)\frac{e^{ikr}}{r},$$

Since

$$\sin(kr - l\pi/2) = [(-i)^l e^{ikr} - i^l e^{-ikr}]/2i$$

Therefore

$$\psi(r,\theta) \longrightarrow -\frac{e^{-ikr}}{2ikr}\sum_{l=0}^{\infty} i^{2l}(2l+1)P_l(\cos\theta) + \frac{e^{ikr}}{r}\left[f(\theta) + \frac{1}{2ik}\sum_{l=0}^{\infty} i^l(-i)^l(2l+1)P_l(\cos\theta)\right]$$

Furthermore, we need first to determine the asymptotic form of the radial function $R_{kl}(r)$. At large values of r, the scattering potential is effectively zero, for it is short range.

$$\left(\frac{d^2}{dr^2} + k^2\right)(r\,R_{kl}(r)) = 0.$$

The general solution of this equation is given by a linear combination of the spherical Bessel and Neumann functions

$$R_{kl}(r) = A_l\, j_l(kr) + B_l n_l(kr),$$

where the asymptotic form of the Neumann function is

$$n_l(kr) \longrightarrow -\frac{\cos(kr - l\pi/2)}{kr} \qquad (r \longrightarrow \infty).$$

We obtain the asymptotic form of the radial function:

$$R_{kl}(r) \longrightarrow A_l \frac{\sin(kr - l\pi/2)}{kr} - B_l \frac{\cos(kr - l\pi/2)}{kr} \qquad (r \longrightarrow \infty).$$

Since the Neumann function diverges at r=0, the cosine term in does not represent a physically acceptable solution; hence, it needs to be discarded near the origin.

$$R_{kl}(r) \longrightarrow C_l \frac{\sin(kr - l\pi/2 + \delta_l)}{kr} \qquad (r \longrightarrow \infty),$$

We see that, with δ_l=0, the radial function $R_{kl}(r)$ is finite at r=0, and reduces to $j_l(\mathrm{k}r)$. So δ_l is a real angle which vanishes for all values of l in the absence of the scattering potential (i.e., V=0); δ_l is called the *phase shift*.

Let us work out a specific example. We consider scattering by a hard, or rigid, sphere

$$V = \begin{cases} \infty & \text{for} \quad r < a \\ 0 & \text{for} \quad r > a. \end{cases}$$

All we need to know is that the wave function must vanish at r = R because the sphere is impenetrable. Therefore, the solution of wave function with l=0 can be written as

$$u(r) = \begin{cases} u_1(r) = 0, & r < a, \\ u_2(r) = A\sin(kr + \delta_0), & r > a, \end{cases}$$

where

$$k^2 = 2mE/\hbar^2$$

The continuity of wave function at r=a

$$\sin(ka+\delta_0)=0 \quad \Longrightarrow \quad \tan\delta_0=-\tan(ka) \quad \Longrightarrow \quad \sin^2\delta_0=\sin^2(ka),$$

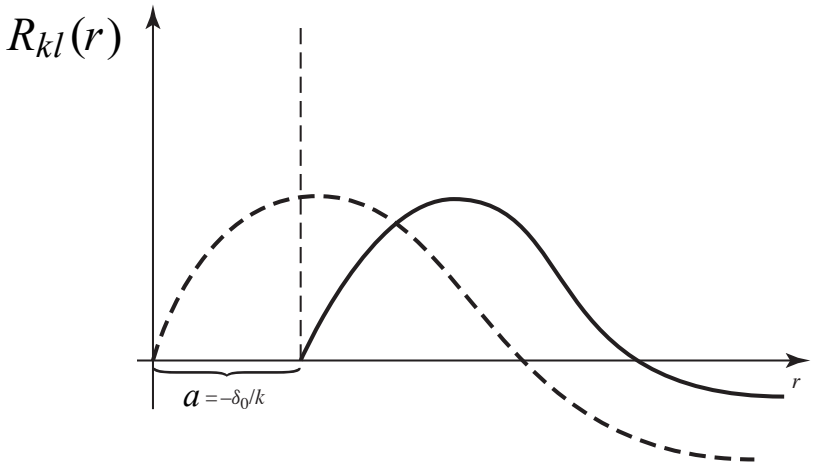
Since

$$\sin^2\alpha=1/(1+\cot^2\alpha).$$

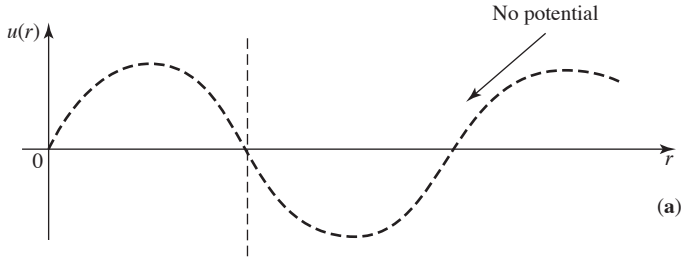
The lowest value of the phase shift is

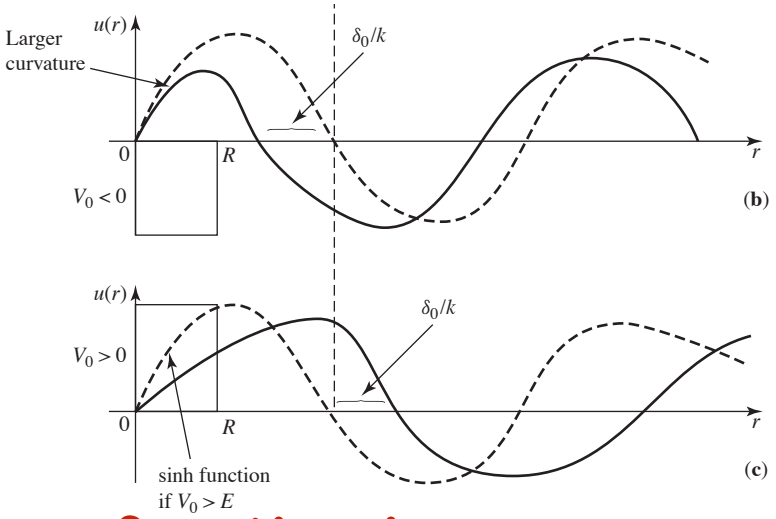
$$\delta_0=-ka$$

It is negative, as it should be for a repulsive potential.



Partial Wave Analysis





The asymptotic limit of wave function is

$$\psi(r,\theta) \longrightarrow \sum_{l=0}^{\infty} a_l P_l(\cos\theta)\frac{\sin(kr - l\pi/2 + \delta_l)}{kr} \qquad (r \longrightarrow \infty).$$

This wave function is known as a *distorted plane wave*, for it differs from a plane wave by having phase shifts and can write as

$$\psi(r,\theta) \longrightarrow -\frac{e^{-ikr}}{2ikr}\sum_{l=0}^{\infty} a_l i^l e^{-i\delta_l} P_l(\cos\theta) + \frac{e^{ikr}}{2ikr}\sum_{l=0}^{\infty} a_l (-i)^l e^{i\delta_l} P_l(\cos\theta).$$

Therefore

$$a_l = (2l+1)i^l e^{i\delta_l}.$$

And

$$f(\theta) + \frac{1}{2ik}\sum_{l=0}^{\infty} i^l(-i)^l(2l+1)P_l(\cos\theta) = \frac{1}{2ik}\sum_{l=0}^{\infty}(2l+1)i^l(-i)^l e^{2i\delta_l}P_l(\cos\theta),$$

Finally, the scattering amplitude can be written as

$$f(\theta) = \sum_{l=0}^{\infty} f_l(\theta) = \frac{1}{2ik}\sum_{l=0}^{\infty}(2l+1)P_l(\cos\theta)(e^{2i\delta_l}-1) = \frac{1}{k}\sum_{l=0}^{\infty}(2l+1)e^{i\delta_l}\sin\delta_l P_l(\cos\theta),$$

We can obtain the differential and the total cross sections

$$\frac{d\sigma}{d\Omega} = \left| f(\theta)\right|^2 = \frac{1}{k^2}\sum_{l=0}^{\infty}\sum_{l'=0}^{\infty}(2l+1)(2l'+1)e^{i(\delta_l-\delta_{l'})}\sin\delta_l\sin\delta_{l'}P_l(\cos\theta)P_{l'}(\cos\theta),$$

Partial Wave Analysis

$$\sigma = \int \frac{d\sigma}{d\Omega} d\Omega = \int_0^{\pi} |f(\theta)|^2 \sin\theta \, d\theta \int_0^{2\pi} d\varphi = 2\pi \int_0^{\pi} |f(\theta)|^2 \sin\theta \, d\theta$$

$$= \frac{2\pi}{k^2} \sum_{l=0}^{\infty} \sum_{l'=0}^{\infty} (2l+1)(2l'+1) e^{i(\delta_l - \delta_{l'})} \sin\delta_l \sin\delta_{l'} \int_0^{\pi} P_l(\cos\theta) P_{l'}(\cos\theta) \sin\theta \, d\theta.$$

Using the relation

$$\int_0^{\pi} P_l(\cos\theta) P_{l'}(\cos\theta) \sin\theta \, d\theta = [2/(2l+1)]\delta_{ll'},$$

the total cross sections are reduced

$$\sigma = \sum_{l=0}^{\infty} \sigma_l = \frac{4\pi}{k^2} \sum_{l=0}^{\infty} (2l+1) \sin^2 \delta_l,$$

where σ_l are called the *partial cross sections* corresponding to the scattering of particles in various angular momentum states.

We should note that in the case where we have a scattering between particles that are in their respective s states, l=0, the scattering amplitude becomes

$$f_0 = \frac{1}{k} e^{i\delta_0} \sin\delta_0 \qquad (l = 0),$$

Since f_0 does not depend on , the differential and total cross sections are given by the following simple relations:

$$\frac{d\sigma}{d\Omega} = |f_0|^2 = \frac{1}{k^2}\sin^2\delta_0, \qquad \sigma = 4\pi\,|f_0|^2 = \frac{4\pi}{k^2}\sin^2\delta_0 \qquad (l = 0).$$

An important issue here is the fact that the total cross section can be related to the *forward* scattering amplitude f(0).

$$f(0) = \frac{1}{k}\sum_{l=0}^{\infty}(2l+1)\left(\sin\delta_l\cos\delta_l + i\sin^2\delta_l\right),$$

The connection between forward scattering amplitude and total cross section is

$$\frac{4\pi}{k}\,\mathrm{Im}\,f(0)=\sigma=\frac{4\pi}{k}\sum_{l=0}^{\infty}(2l+1)\sin^2\delta_l.$$

This is known as the *optical theorem* (it is reminiscent of a similar theorem in optics which deals with the scattering of light). The physical origin of this theorem is the conservation of particles (or probability). The beam emerging (after scattering) along the incidence direction (θ=0) contains fewer particles than the incident beam, since a number of particles have scattered in various directions.

The scattering amplitude can be rewritten as

$$f(\theta) = \sum_{l=0}^{\infty} (2l+1) f_l(k) P_l(\cos\theta),$$

where

$$f_l(k) = \frac{1}{k} e^{i\delta_l} \sin\delta_l = \frac{1}{2ik}\left(e^{2i\delta_l} - 1\right) = \frac{1}{2ik}\left(S_l(k) - 1\right),$$

with

$$S_l(k) = e^{2i\delta_l}.$$

In the case where there is no flux loss, we must have $|S_l(k)|=1$. However, this requirement is not valid whenever there is *absorption* of the incident beam.

In this case of flux loss, $S_l(k)$ is redefined by

$$S_l(k) = \eta_l(k) e^{2i\delta_l},$$

hence

$$f_l(k) = \frac{\eta_l e^{2i\delta_l} - 1}{2ik} = \frac{1}{2k}\left[\eta_l \sin 2\delta_l + i(1 - \eta_l \cos 2\delta_l)\right].$$

The scattering amplitude is

$$f(\theta) = \frac{1}{2k}\sum_{l=0}^{\infty}(2l+1)\left[\eta_l \sin 2\delta_l + i(1 - \eta_l \cos 2\delta_l)\right] P_l(\cos\theta).$$

The *total elastic* scattering cross section is given by

$$\sigma_{el} = 4\pi \sum_{l=0}^{\infty}(2l+1)|f_l|^2 = \frac{\pi}{k^2}\sum_{l}(2l+1)(1 + \eta_l^2 - 2\eta_l \cos 2\delta_l).$$

The *total inelastic* scattering cross section, which describes the loss of flux, is given by

$$\sigma_{inel} = \frac{\pi}{k^2}\sum_{l=0}^{\infty}(2l+1)\left(1 - \eta_l^2(k)\right).$$

Thus, if $\eta_l(k)$=1 there is no inelastic scattering, but if $\eta_l(k)$=0 we have total absorption, although there is still elastic scattering in this partial wave.

The total cross section

$$\sigma_{tot} = \sigma_{el} + \sigma_{inel} = \frac{2\pi}{k^2}\sum_{l=0}^{\infty}(2l+1)\left(1-\eta_l\cos(2\delta_l)\right).$$

The optical theorem is also valid for inelastic scattering.

$$\mathrm{Im}\, f(0) = \sum_{l=0}^{\infty}(2l+1)\,\mathrm{Im}\, f_l = \frac{1}{2k}\sum_{l=0}^{\infty}(2l+1)\left(1-\eta_l\cos(2\delta_l)\right)$$

and

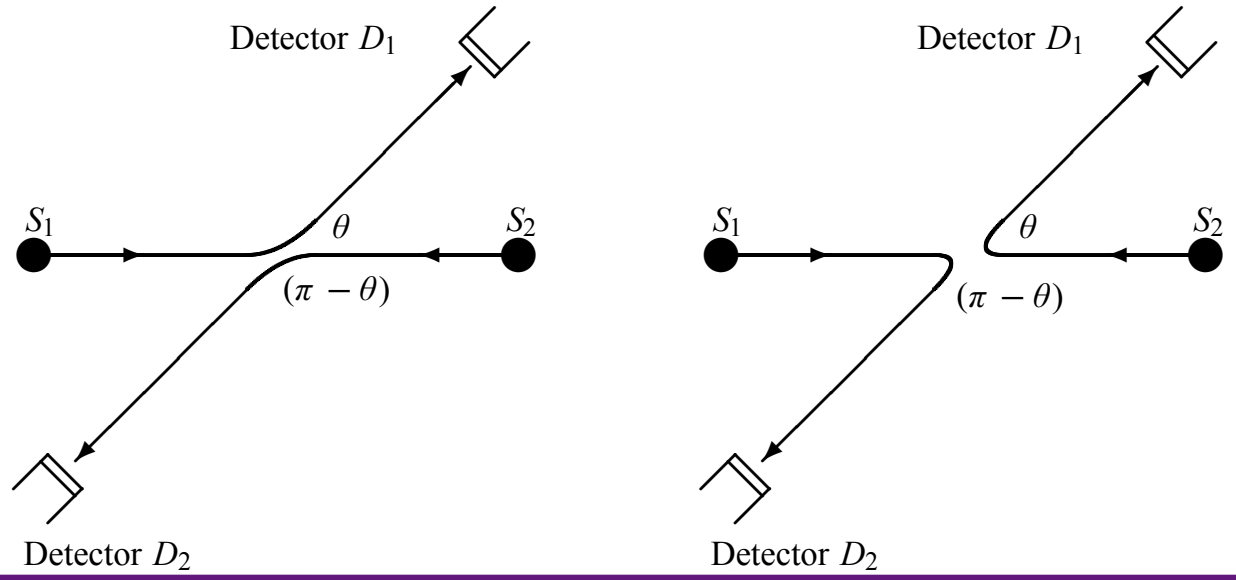
$$\mathrm{Im}\, f(0) = k\sigma_{tot}/4\pi$$

Scattering of Identical Particles

First, let us consider the scattering of two identical *bosons* in their center of mass frame. *Classically*, the cross section for the scattering of two identical particles whose interaction potential is central is given by

$$\sigma_{cl}(\theta) = \sigma(\theta) + \sigma(\pi - \theta).$$

In quantum mechanics there is no way of distinguishing for them



Scattering of Identical Particles

Thus, the scattered wave function must be symmetric:

$$\psi_{sym}(\vec{r}) \longrightarrow e^{i\vec{k}_0\cdot\vec{r}} + e^{-i\vec{k}_0\cdot\vec{r}} + f_{sym}(\theta)\frac{e^{ikr}}{r},$$

and so must also be the scattering amplitude:

$$f_{boson}(\theta) = f(\theta) + f(\pi-\theta).$$

Therefore, the differential cross section is

$$\frac{d\sigma}{d\Omega}_{boson} = \Big|f(\theta)+f(\pi-\theta)\Big|^2 = \Big|f(\theta)\Big|^2 + \Big|f(\pi-\theta)\Big|^2 + f(\theta)^* f(\pi-\theta) + f(\theta)f^*(\pi-\theta)$$

$$= \Big|f(\theta)\Big|^2 + \Big|f(\pi-\theta)\Big|^2 + 2\,\mathrm{Re}\Big[f^*(\theta)f(\pi-\theta)\Big].$$

When $\theta=\pi/2$,

$$(d\sigma/d\Omega)_{boson} = 4\,|f(\pi/2)|^2$$

Consider now the scattering of two identical spin 1/2 particles. The wave function of a two spin 1/2 particle system is known to be either symmetric or antisymmetric. When the spatial wave function is symmetric, that is the two particles are in a spin singlet state, the differential cross section is given by

$$\frac{d\sigma_S}{d\Omega} = |f(\theta) + f(\pi - \theta)|^2 ,$$

but when the two particles are in a spin triplet state, the spatial wave function is antisymmetric, and hence

$$\frac{d\sigma_A}{d\Omega} = |f(\theta) - f(\pi - \theta)|^2 .$$

If the incident particles are unpolarized, the various spin states will be equally likely, so the triplet state will be three times as likely as the singlet:

$$\frac{d\sigma}{d\Omega}_{fermion} = \frac{3}{4}\frac{d\sigma_a}{d\Omega} + \frac{1}{4}\frac{d\sigma_s}{d\Omega} = \frac{3}{4}|f(\theta) - f(\pi-\theta)|^2 + \frac{1}{4}|f(\theta) + f(\pi-\theta)|^2$$
$$= |f(\theta)|^2 + |f(\pi-\theta)|^2 - \mathrm{Re}\left[f^*(\theta)f(\pi-\theta)\right].$$

We should note that, in the case of partial wave analysis for elastic scattering

$$f(\pi-\theta) = \frac{1}{k}\sum_{l=0}^{\infty}(2l+1)e^{i\delta_l}\sin\delta_l P_l(\cos(\pi-\theta))$$
$$= \frac{1}{k}\sum_{l=0}^{\infty}(-1)^l(2l+1)e^{i\delta_l}\sin\delta_l P_l(\cos\theta),$$

and

$$f(\theta) \pm f(\pi-\theta) = \frac{1}{k}\sum_{l=0}^{\infty}\left[1 \pm (-1)^l\right](2l+1)e^{i\delta_l}\sin\delta_l P_l(\cos\theta).$$

Calculate the total cross section for the low-energy scattering of a particle of mass m from an attractive square well potential $V(r)=-V_0$ for $r<a$ and $V(r)=0$ for $r>a$, with $V_0>0$.

Solution:

Since the scattering is dominated at low energies by the s partial waves, $l=0$, the Schrödinger equation for the radial function is given by

$$-\frac{\hbar^2}{2m}\frac{d^2u(r)}{dr^2} - V_0 u(r) = Eu(r) \qquad (r < a),$$

$$-\frac{\hbar^2}{2m}\frac{d^2u(r)}{dr^2} = Eu(r) \qquad (r > a),$$

where

$$u(r) = r\,R(r).$$

The solutions of these equations for positive energy states are

$$u(r) = \begin{cases} u_1(r) = A\sin(k_1 r), & r < a, \\ u_2(r) = B\sin(k_2 r + \delta_0), & r > a, \end{cases}$$

and

$$k_1^2 = 2m(E + V_0)/\hbar^2 \text{ and } k_2^2 = 2mE/\hbar^2.$$

The continuity of $u(r)$ and its first derivative at r=a

$$\left.\frac{u_2(r)}{u_2'(r)}\right|_{r=a} = \left.\frac{u_1(r)}{u_1'(r)}\right|_{r=a} \quad \Longrightarrow \quad \frac{1}{k_2}\tan(k_2 a + \delta_0) = \frac{1}{k_1}\tan(k_1 a),$$

which yields

$$\delta_0 = -k_2 a + \tan^{-1}\left[\frac{k_2}{k_1}\tan(k_1 a)\right].$$

Since

$$\tan(k_2 a + \delta_0) = \frac{\sin(k_2 a)\cos\delta_0 + \cos(k_2 a)\sin\delta_0}{\cos(k_2 a)\cos\delta_0 - \sin(k_2 a)\sin\delta_0} = \frac{\tan(k_2 a) + \tan\delta_0}{1 - \tan(k_2 a)\tan\delta_0},$$

therefore,

$$\tan\delta_0 = \frac{k_2\tan(k_1 a) - k_1\tan(k_2 a)}{k_1 + k_2\tan(k_1 a)\tan(k_2 a)}.$$

Using the relation

$$\sin^2\delta_0 = 1/(1 + 1/\tan^2\delta_0),$$

we can write

$$\sin^2\delta_0 = \left[1 + \left(\frac{k_1 + k_2\tan(k_1 a)\tan(k_2 a)}{k_2\tan(k_1 a) - k_1\tan(k_2 a)}\right)^2\right]^{-1},$$

So

$$\sigma_0 = \frac{4\pi}{k_1^2}\sin^2\delta_0 = \frac{4\pi}{k_1^2}\left[1 + \left(\frac{k_1 + k_2\tan(k_1 a)\tan(k_2 a)}{k_2\tan(k_1 a) - k_1\tan(k_2 a)}\right)^2\right]^{-1}.$$